The Hebrew University of Jerusalem

Department of Physics

Final project

N-body gravitational simulation

Barnes-Hut tree with adaptive RK4 integration

Eran Rehani

Course 77315: Computational Physics

2026

Contents

1 Problem and initial conditions

The simulations evolve $N = 5000$ equal mass particles with total mass $M = 1 \times 10^{11}\, M_\odot$ for 20 Gyr. Their initial positions have uniform volume density inside a sphere of radius $R = 50\,\text{kpc}$. The position sampler uses $r = Ru^{1/3}$ for $u \sim U(0,1)$ and chooses an isotropic direction from $\cos\vartheta = 2u - 1$ and $\varphi = 2\pi u$.

Each velocity component follows a normal distribution with $\sigma = V/\sqrt{3}$, so $V = \langle v^2 \rangle^{1/2}$. The three runs use $V = 65$, 80, and 95 km/s. All three reuse the same positions and standardized velocity draw; only the velocity scale changes. The initialization routine subtracts the sampled mean velocity but leaves the positions unchanged, keeping every particle inside the assigned sphere.

The assignment uses the softened force

$$\mathbf{f}_{ij} = -\frac{Gm_i m_j}{(r_{ij} + S)^2} \frac{\mathbf{r}_{ij}}{r_{ij}}, \qquad \mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j, \tag{1}$$

with $S = 1\,\text{kpc}$. Its pair potential, $U_{ij} = -Gm_i m_j/(r_{ij} + S)$, defines the energy diagnostic. After each accepted step, the solver removes particles with any coordinate outside the ± 666 kpc calculation box.

2 Numerical method

2.1 Units

The code uses kpc, $M_\odot$, and Gyr. In these units,

$$G = 4.4996 \times 10^{-6}\,\text{kpc}^3\,\text{M}_\odot^{-1}\,\text{Gyr}^{-2}, \qquad 1\,\text{kpc/Gyr} = 0.977\,792\,2\,\text{km/s}. \tag{2}$$

Each simulation particle has mass $m = M/N = 2 \times 10^7\, M_\odot$.

2.2 Barnes-Hut tree

The tree method follows Barnes and Hut [1]. At each force evaluation, the tree builder places the particles in a cubic root cell and splits every cell containing more than one particle into eight octants. Each cell stores its particle count, mass, center of mass, and main diagonal $L = \sqrt{3}\,l$. The brief defines L as the diagonal, rather than the side length l used in the usual opening criterion.

For each target particle, the tree walk opens any cell that contains it and evaluates a leaf by direct softened interaction, excluding self-interaction. For an external cell at center-of-mass distance D, it recurses if $D/L < \theta$ and replaces the cell by a point mass if $D/L \geq \theta$. The solver uses the assigned $\theta = 1$ and rebuilds the tree at every Runge-Kutta stage.

Dividing Equation 1 by the target mass gives one source's acceleration,

$$\mathbf{a}_{ij} = -\frac{Gm_j}{(r_{ij} + S)^2} \frac{\mathbf{r}_{ij}}{r_{ij}}. \tag{3}$$

Small tests use a direct $O(N^2)$ calculation as the reference. Figure 1 illustrates the same rules with a two-dimensional quadtree. For this 60-particle example, the target requires 12 direct pairs and 14 cell terms instead of 59 direct pairs.

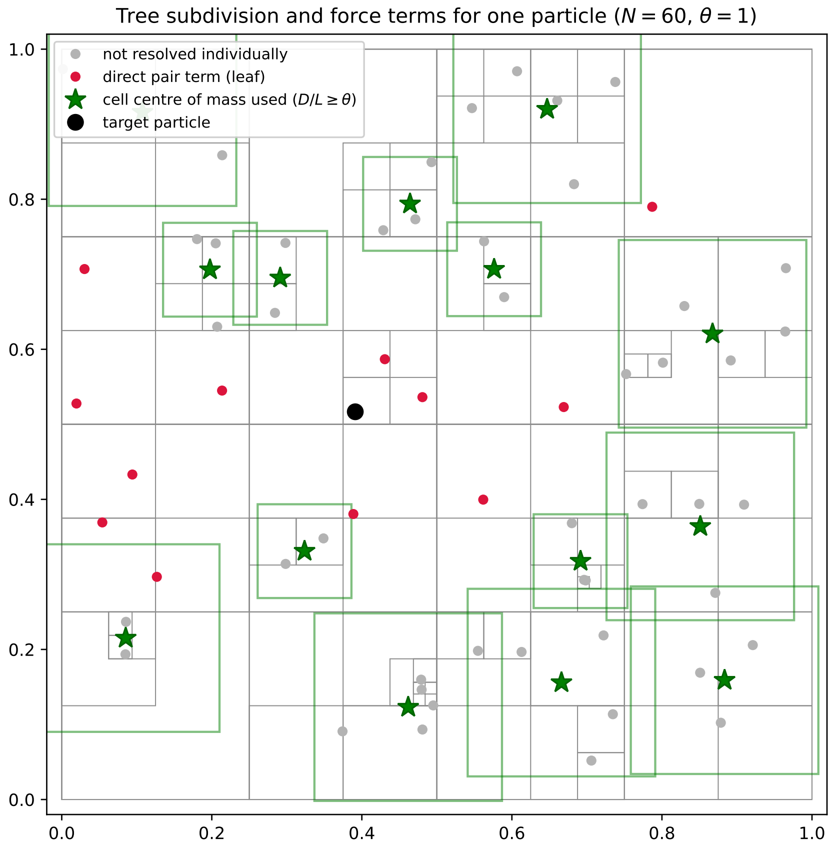

Figure 1: Two-dimensional illustration of the tree method with $\theta = 1$. Thin lines are cell boundaries. Red points contribute a direct pair force to the black target particle. Green stars are cell centers of mass that satisfy $D/L \geq \theta$, drawn inside the cell they replace. Grey points are absorbed into those cells.

2.3 Adaptive RK4 integration

The solver represents the state as $\mathbf{y} = (\mathbf{r}_1, \ldots, \mathbf{r}_N, \mathbf{v}_1, \ldots, \mathbf{v}_N)$ and integrates

$$\dot{\mathbf{r}}_i = \mathbf{v}_i, \qquad \dot{\mathbf{v}}_i = \mathbf{a}_i. \tag{4}$$

For a proposed interval h, the controller compares one fourth-order Runge-Kutta step with two steps of length $h/2$. The comparison costs twelve force evaluations. The brief defines the error as the largest particle displacement difference,

$$\text{err} = \max_i \left\| \mathbf{r}_{i,\text{half}} - \mathbf{r}_{i,\text{full}} \right\|. \tag{5}$$

The controller accepts the step when err $\leq$ tol and applies Richardson extrapolation,

$$\mathbf{y}_{\text{acc}} = \mathbf{y}_{\text{half}} + \frac{\mathbf{y}_{\text{half}} - \mathbf{y}_{\text{full}}}{2^4 - 1}. \tag{6}$$

The controller multiplies h by $0.9(\text{tol}/\text{err})^{1/5}$, with a maximum growth factor of two and a minimum reduction factor of 0.1. It caps the step between 0.01 and 2 Gyr. If the controller reaches the lower bound and still cannot meet the tolerance, the code stops with an error instead of accepting a state computed for a different step length.

The brief gives the displacement accuracy in units of the box size, $\varepsilon = 0.1$, which gives

$$\text{tol} = \varepsilon L_{\text{box}} = 0.1 \times 1332\,\text{kpc} = 133.2\,\text{kpc}, \tag{7}$$

a permitted step error of 133 kpc for an initial cluster radius of 50 kpc. Section 4 tests this value.

The integrator clips steps at 0, 2.5, 5, 10, 15, and 20 Gyr so each stored snapshot matches its label.

2.4 Energy and escapers

At every accepted step, the energy diagnostic calculates

$$E_k = \frac{1}{2} \sum_i m_i v_i^2, \qquad E_g = \frac{1}{2} \sum_i m_i \Phi_i, \qquad \Phi_i = -G \sum_{j \neq i} \frac{m_j}{r_{ij} + S}. \tag{8}$$

For $N > 1500$, the diagnostic evaluates Φ_i with the force tree; smaller tests use the direct sum. When particles leave, the bookkeeping stores their kinetic energy and calculates E_g before and after removal. The difference includes escaper-escaper and escaper-retained terms within the chosen estimator. At production size, separate Barnes-Hut approximations produce the two values. The retained diagnostic contains active particles; the corrected diagnostic adds the required escape values. Neither includes later interactions because removed particles no longer evolve.

2.5 Density profile and King fit

The density analysis measures radius from the retained center of mass, which moves by a few kpc after escaper removal, and divides the occupied range into 25 logarithmic shells. For shell edges r_j and r_{j+1},

$$\rho_j = \frac{M_j}{\frac{4\pi}{3}(r_{j+1}^3 - r_j^3)}, \qquad r_{j,\text{plot}} = \sqrt{r_j r_{j+1}}. \tag{9}$$

For equal mass particles, the Poisson uncertainty is $\delta\rho_j = \rho_j/\sqrt{N_j}$. The implementation also supports unequal masses by summing m_i^2 in each shell.

The fit uses the model requested in the assignment,

$$\rho(r) = \frac{\rho_c}{[1 + (r/r_c)^\alpha]^\beta}. \tag{10}$$

The analysis fits $\log\rho$ with propagated error $\delta\log\rho \simeq \delta\rho/\rho$. Optimizing $\log\rho_c$ and $\log r_c$ keeps both parameters positive. The $V = 80$ fit scans several starting values for α and β, then holds them fixed while fitting ρ_c and r_c for $V = 65$ and $V = 95$.

3 Implementation and checks

The submitted `nbody_solver.py` contains the sampler, tree walk, adaptive integrator, diagnostics, fits, plots, and self-tests. Numba compiles a flattened tree and distributes target particles across threads; the Python walk provides a slower fallback. The implementation adapts the RK controller from HW_05 to use the per-particle norm in Equation 5 and to resize the state after escapes. The isotropic direction sampler comes from HW_07.

The command `python nbody_solver.py --test` runs the checks, and `--tol=X` runs all three $N = 5000$ cases. The solver stores states, energy histories, escaper records, figures, fit tables, and numerical summaries under `figures/`. Its redraw and refit modes reuse these states without repeating the dynamics. A separate `nbody_solver.py` file accompanies this report.

In the self-test, 5000 samples give a maximum radius of 49.9905 kpc and an RMS speed of 79.422 km/s for the 80 km/s distribution. The production draw gives 49.9961 kpc and 80.235 km/s. These sub-percent deviations match sampling noise, and subtracting the mean sets the center-of-mass velocity to machine zero.

For 150 particles with unequal masses, the Barnes-Hut acceleration at $\theta = 1$ had an L_2 relative error of 1.014×10^{-2} against the direct sum. The tree gravitational energy differed from the direct result by 5.214×10^{-4}. A softened unequal-mass circular binary completed two periods with a relative energy change of 1.29×10^{-10} under a tight tolerance, and its final separation agreed with the initial value to the reported precision. A separate constant-velocity test confirmed that stored snapshots match their labels.

The brief's opening convention differs from the standard side-length form. Since $L = \sqrt{3}\,l$,

$$\frac{D}{L} \geq \theta_{\text{diagonal}} \quad \Longleftrightarrow \quad \frac{l}{D} \leq \frac{1}{\sqrt{3}\,\theta_{\text{diagonal}}} \equiv \theta_{\text{side}}. \tag{11}$$

Table 1 compares the tree acceleration with a direct sum for 3000 particles. At the assigned $\theta = 1$, the L_2 error is 9.14×10^{-3}, consistent with the smaller random test.

Raising θ to 2 cuts the error by a factor of eight. One force evaluation then takes 96 ms instead of 89 ms because tree construction, which does not depend on θ, dominates at $N = 5000$. The production runs retain the assigned value.

The production controller accepted 310, 262 and 352 steps for $V = 80$, 95 and 65 km/s, with 4, 10 and 3 rejected proposals. The runs took 1155 s in total on a twenty-core machine. The assigned tolerance produced no rejections in the corresponding full runs.

Table 1: Barnes-Hut acceleration error against direct summation for the initial uniform sphere, $N = 3000$.

θ (diagonal)	θ (side)	L_2 relative error in $\mathbf{a}$
2.0	0.289	1.17e-03
1.5	0.385	3.07e-03
1.0	0.577	9.14e-03
0.8	0.722	1.58e-02
0.6	0.962	3.55e-02
0.4	1.443	1.03e-01

4 Choice of integration tolerance

The assigned tolerance, 133.2 kpc, exceeds the initial cluster radius. The tolerance study repeats the $V = 80$ km/s run at four values across three decades. For each run, the energy diagnostic measures the retained particle energy plus the escape values frozen at removal. This quantity removes deletion jumps but is not exact: the Barnes-Hut force and tree energy both introduce spatial approximation. Changing the RK tolerance isolates the timestep-dependent contribution.

Table 2: Energy drift versus RK4 position tolerance for $V = 80$ km/s, $N = 5000$, 20 Gyr, including frozen escape values.

tol [kpc]	steps	rejected	N_f	$\max \lvert\Delta E/E_0\rvert$
133.2	20	1	4781	4.82e-01
13.32	33	0	4833	8.37e-02
1.332	97	0	4829	1.77e-02
0.1332	303	4	4833	2.53e-03

Tightening the tolerance to 0.1332 kpc lowers the worst energy excursion from 48 percent to 0.25 percent while increasing the step count from 20 to 303. Section 5 uses this value. Appendix A contains all required outputs at the assigned value; those files use the suffix `_eps01`.

Reducing the tolerance by 10^3 cuts the worst excursion by a factor of 190. The one-percent force error at $\theta = 1$ and temporally unresolved close encounters set the remaining floor.

5 Results

5.1 Initial energies

For an unsoftened uniform sphere, $E_g = -3GM^2/(5R)$. Since $E_k = MV^2/2$, the virial value $2E_k/\lvert E_g\rvert = 1$ occurs at

$$V_{\mathrm{vir}} = \sqrt{\frac{3GM}{5R}} = 71.85\,\mathrm{km/s}. \tag{12}$$

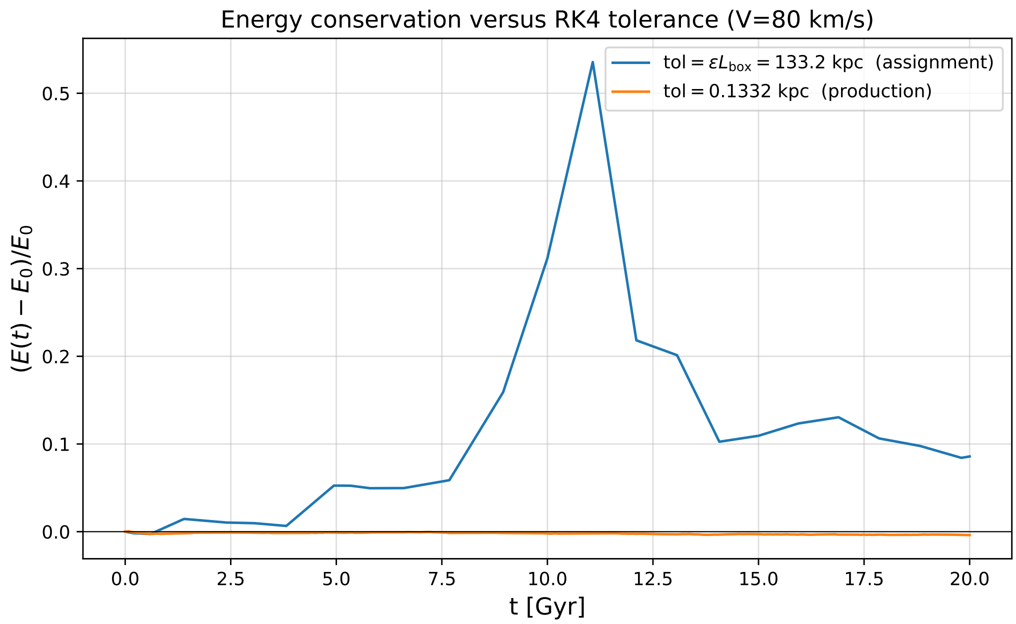

Figure 2: Relative energy change for $V = 80\,\mathrm{km/s}$ at the tolerance of the assignment and at the tolerance used for the production runs.

The $V = 65$ case starts below virial equilibrium, while the $V = 80$ and $V = 95$ cases start above it. All three have $2E_k/|E_g| < 2$ and negative initial total energy. The $V = 95$ state is close to zero energy, but it is not initially unbound.

Table 3: Initial and final run data. Energies use the simulation units.

V [km/s]	$2E_{k,0}/\|E_{g,0}\|$	E_0 [$10^{14} M_\odot\,\mathrm{kpc}^2\,\mathrm{Gyr}^{-2}$]	removed	retained	steps
65	0.849	−3.013	20	4980	352
80	1.286	−1.868	167	4833	310
95	1.814	−0.488	758	4242	262

5.2 (a) Particle positions

Figure 3 shows the $V = 80$ positions at the six requested times. The panels frame the cluster because the full box would hide its structure. By 2.5 Gyr, the uniform sphere has formed a dense core and an extended halo. The halo expands throughout the run, and 167 particles cross the box boundary.

5.3 (b) Energy conservation

The curves in Figure 4 separate after the first escape at 5.3 Gyr. At 20 Gyr, the retained energy gives +0.034 while the frozen-escape accounting gives −0.0039, also its largest excursion. Removal deletes negative binding terms and creates most of the apparent gain in the retained curve. At $V = 95$, where 758 particles escape, the two final values are +0.81 and +0.0013. At $V = 65$, only 20 leave and both values end near −0.003.

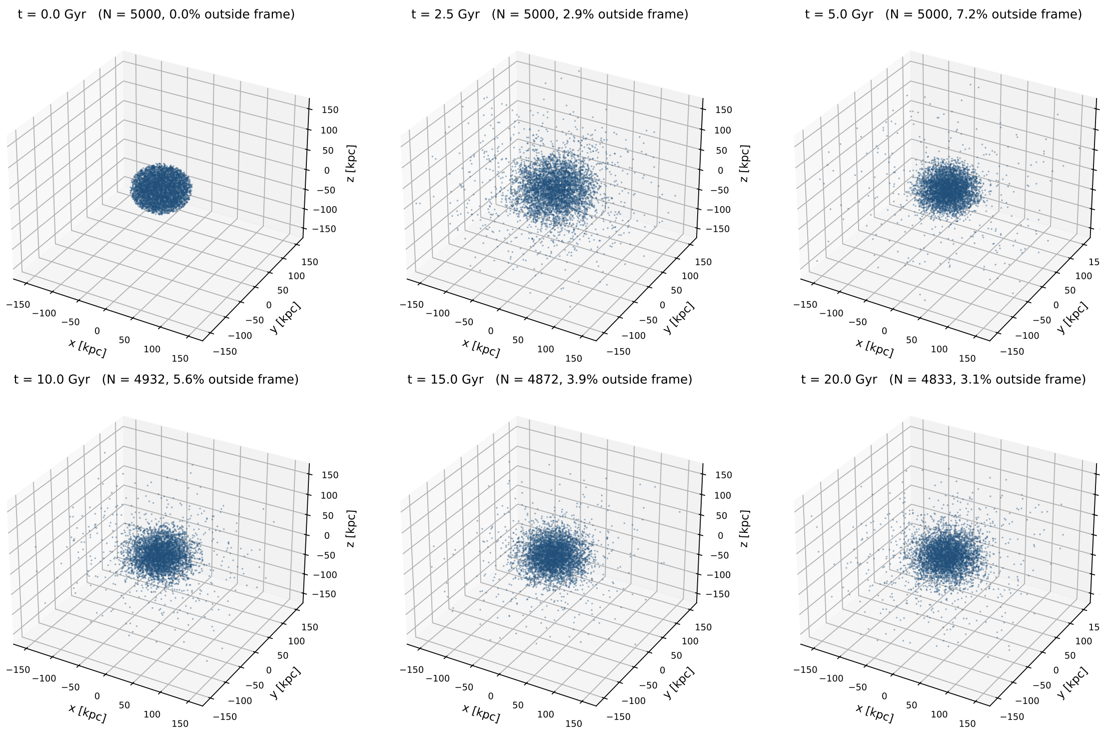

Figure 3: Particle positions for the $V = 80\,\mathrm{km/s}$ run. Each panel shows an integrated state at the time printed above it.

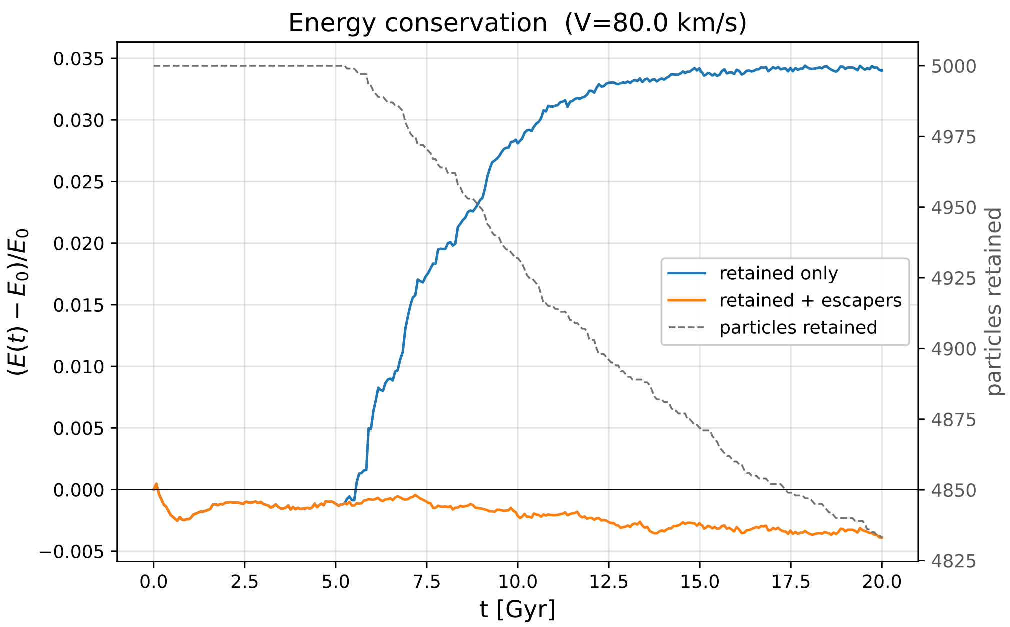

Figure 4: Relative energy change for $V = 80\,\mathrm{km/s}$. The blue curve counts only the particles still inside the box. The orange curve adds back the kinetic and potential energy each escaper carried away at the moment it was removed. The dashed grey line, on the right axis, is the number of particles still being integrated.

5.4 (c) Virial ratio

The $V = 80$ ratio starts at 1.286, falls to 0.82 by 2.5 Gyr, and then oscillates about one with decreasing amplitude (Figure 5). Maxima at 5.1, 9.4, 14.0 and 18.6 Gyr give a 4.5 Gyr period. The final retained value is 0.954. The expansion, loss of fast particles and damped oscillation toward unity are characteristic of violent relaxation.

The final retained ratios are 0.960, 0.954 and 0.937 for $V = 65$, 80 and 95 km/s. The $V = 65$ system starts at 0.849, contracts, and loses 20 particles. The $V = 95$ system starts at 1.814 and loses 15 percent of its mass as it approaches equilibrium. Its two escape-accounting curves differ most.

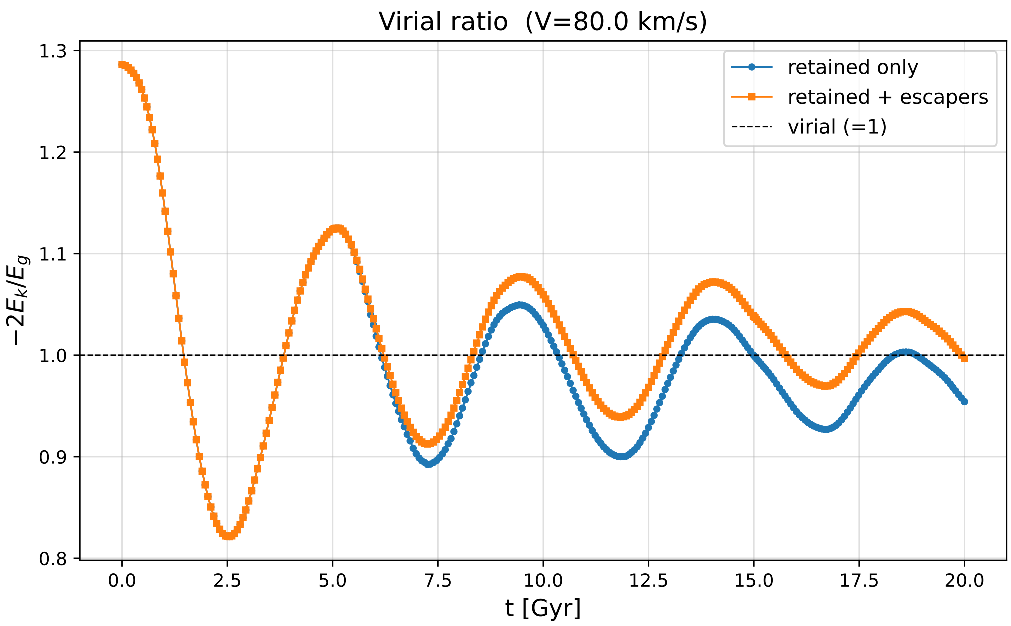

Figure 5: Virial ratio for $V = 80$ km/s. The horizontal line marks $-2E_k/E_g = 1$.

5.5 (d) Density profiles and King fits

Figure 6 shows a monotonic trend with initial velocity. The core radius grows from 24.7 kpc at $V = 65$ through 35.1 kpc at $V = 80$ to 57.4 kpc at $V = 95$, and the central density falls by more than a factor of ten across the same range, from 7.5×10^5 to $5.4 \times 10^4\ M_\odot\ \mathrm{kpc}^{-3}$. Higher initial velocity leaves a larger, more diffuse remnant.

Table 4: King-model parameters. The uncertainties are the formal one standard deviation errors from the weighted fit.

V [km/s]	ρ_c [M$_\odot$/kpc^3]	r_c [kpc]	α	β
80	2.455e+05 ± 1.09e+04	35.125 ± 0.447	4.831 ± 0.385	0.986 ± 0.085
95	5.380e+04 ± 1.59e+03	57.339 ± 0.506	4.831 (fixed)	0.986 (fixed)
65	7.469e+05 ± 2.16e+04	24.673 ± 0.211	4.831 (fixed)	0.986 (fixed)

The common shape parameters are $\alpha = 4.83 \pm 0.38$ and $\beta = 0.986 \pm 0.085$. The outer profile constrains their product because $\rho \propto r^{-\alpha\beta}$ at large radius. Here $\alpha\beta = 4.77$, close to

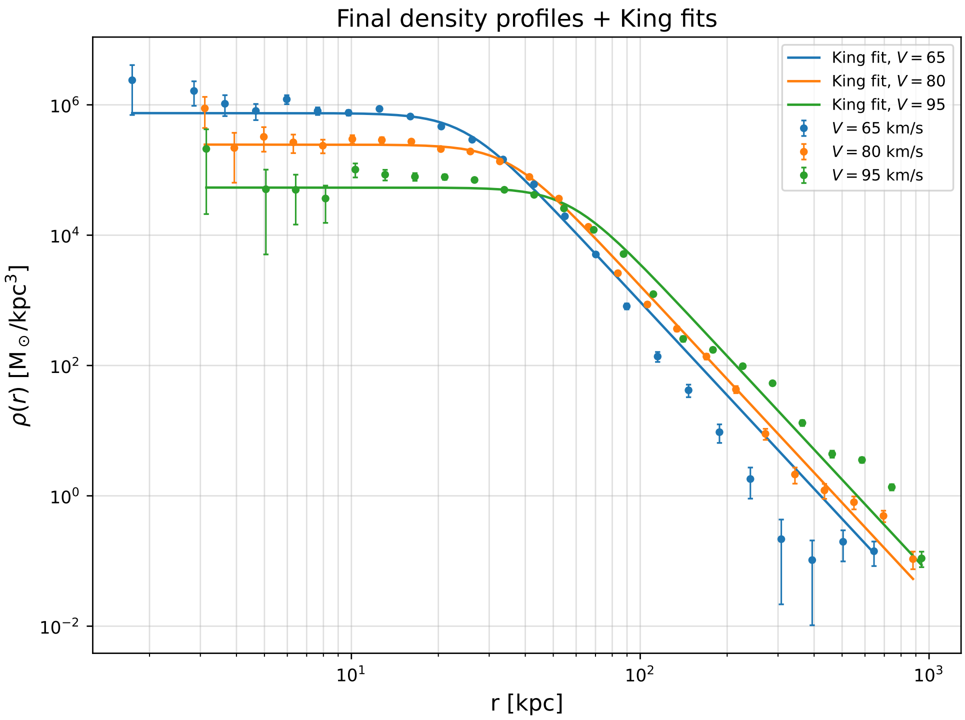

Figure 6: Center of mass density profiles at 20 Gyr with King-model fits. The $V = 65$ and $V = 95$ fits use the α and β values obtained from $V = 80$.

the Plummer value of 5. The assigned-tolerance states give $\alpha = 0.700$, $\beta = 7.389$, and a similar product of 5.17. The data therefore constrain the outer slope more strongly than either shape parameter.

The reduced χ^2 values, 7.7, 44.7 and 12.8 for $V = 80$, 95 and 65 km/s, exceed unity. Populated bins have small Poisson errors, so departures from the King form dominate the statistic; the $V = 95$ profile also has a shallow central depression that a monotonic King curve cannot fit. The quoted uncertainties describe local parameter curvature, not model mismatch.

6 Numerical difficulties

The assigned tolerance is too loose to control the integration: it exceeds the system size and permits energy errors of tens of percent (Section 4). RK4 is also non-symplectic, so it does not have the bounded long-time energy error typical of leapfrog methods. Smaller steps reduce its finite-time error, but the Barnes-Hut force and energy approximations set the residual floor in Table 2.

Close encounters raise the acceleration and can drive the step to its 0.01 Gyr floor. Softening bounds the pair force by Gm_im_j/S^2; if the error remains too large at the floor, the code stops rather than accepting an uncontrolled step. Exact coordinate duplicates require a separate tree-build guard because repeated octant subdivision cannot separate them. The tree stores such particles in one childless cell. They exert zero mutual force and act as one point mass externally, so this treatment is exact. The chosen initial sampler does not generate duplicates.

Computing E_g with the $\theta = 1$ tree walk costs $O(N \log N)$ at every accepted step. Its relative bias is 5.2×10^{-4} against the direct sum for 150 particles, below the measured drift; runs below 1500 particles use the exact sum. Escaper removal itself causes jumps in the retained-particle energy, because it deletes both the particle's kinetic energy and its pair terms. The corrected curves in Figures 4 and 5 separate these jumps from integration error.

For the density fit, 25 logarithmic shells resolve the core better than a linear grid; empty bins are omitted and the remaining points use Poisson weights. Because the outer model depends mainly on $\alpha\beta$, the log-space fit tries several starting points and retains the lowest χ^2. Fixing the fitted shape for the other two velocities removes that degeneracy as required.

Adaptive step doubling needs twelve force evaluations per accepted step, each with a new tree. Flattened tree arrays, a Numba-compiled walk and threaded target-particle loops reduce an $N = 5000$ run from impractical pure Python time to a few minutes.

7 Conclusion

The solver includes the assigned initial conditions, softened force, Barnes-Hut criterion, adaptive RK4 and boundary removal. Direct-sum comparisons check the tree, while softened-binary and snapshot-time tests check the integrator. All three systems approach $-2E_k/E_g \approx 1$ within 20 Gyr: the 65 km/s system contracts and retains almost all its mass, while the 80 km/s and 95 km/s systems expand and lose about 3 and 15 percent. Their fitted core radii increase and central densities decrease with initial speed.

At the assigned 133.2 kpc tolerance, final escape-corrected energy shifts range from 9 to 35 percent. Reducing it by 10^3 keeps all three below 0.5 percent, at the cost of 310 rather than 28 accepted steps for the full-output 80 km/s case. The main results therefore use the converged tolerance; Appendix A retains every requested output at the assigned value.

A Results at the assigned tolerance

This appendix repeats all three runs with the assigned tol $= \varepsilon L_{\text{box}} = 133.2$ kpc instead of the converged 0.1332 kpc. The files use the `_eps01` suffix and the command

```
python nbody_solver.py --tol=133.2 --tag=eps01.
```

The initial conditions, θ, S, box, and end time are unchanged.

The controller accepts 28, 15 and 34 steps for $V = 80$, 95 and 65 km/s, with no rejections. The convergence study in Table 2 uses a smaller step floor and no intermediate snapshot times, so its accepted sequence and drift differ slightly. At both settings the tolerance is larger than the system and the controller provides little control.

The core-halo structure survives at the assigned tolerance, but the corrected energy error reaches 54 percent for the full-output $V = 80$ run. The 48 percent maximum in Table 2 differs because its step floor and output schedule change the accepted sequence. The virial ratio still ends near unity, 0.929 instead of 0.954, but 28 samples give only about six points per oscillation and cannot resolve the measured 4.5 Gyr period.

The density profiles are more seriously distorted. Table 6 gives $r_c = 24.5$, 25.2 and 40.0 kpc for $V = 65$, 80 and 95 km/s; the first two agree within error, erasing the monotonic

Table 5: Assigned and converged runs. The drift is the largest excursion of the escaper-corrected total energy.

	tol = 133.2 kpc (assigned)			tol = 0.1332 kpc (used)		
V [km/s]	steps	retained	$\Delta E/E_0$	steps	retained	$\Delta E/E_0$
65	34	4755	-0.347	352	4980	-0.003
80	28	4822	$+0.086$	310	4833	-0.004
95	15	4239	$+0.219$	262	4242	$+0.001$

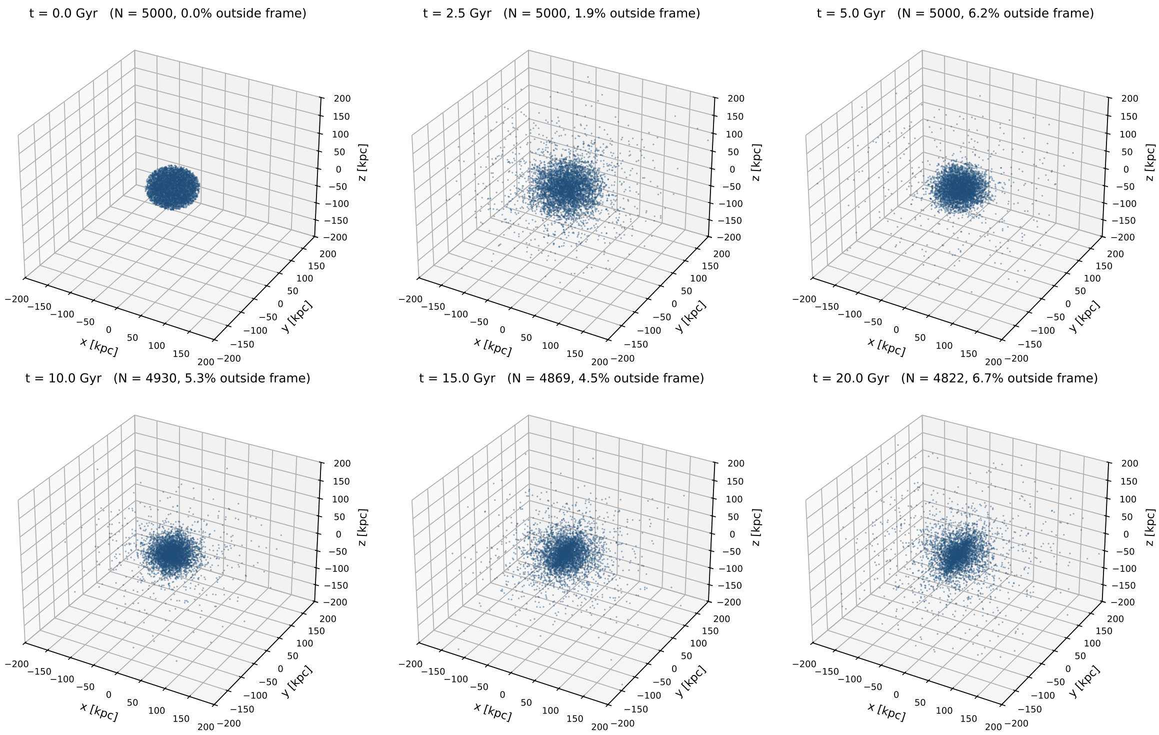

Figure 7: Assigned-tolerance particle positions for $V = 80$ km/s, requirement (a).

Table 6: Assigned-tolerance King parameters. The $V = 80$ fit supplies α and β for the other velocities.

V [km/s]	ρ_c [$M_\odot$/kpc^3]	r_c [kpc]	α	β
80	2.642e+07 ± 6.36e+06	25.217 ± 3.541	0.700 ± 0.050	7.389 ± 0.829
95	6.504e+06 ± 4.26e+05	39.954 ± 0.843	0.700 (fixed)	7.389 (fixed)
65	2.872e+07 ± 1.33e+06	24.525 ± 0.363	0.700 (fixed)	7.389 (fixed)

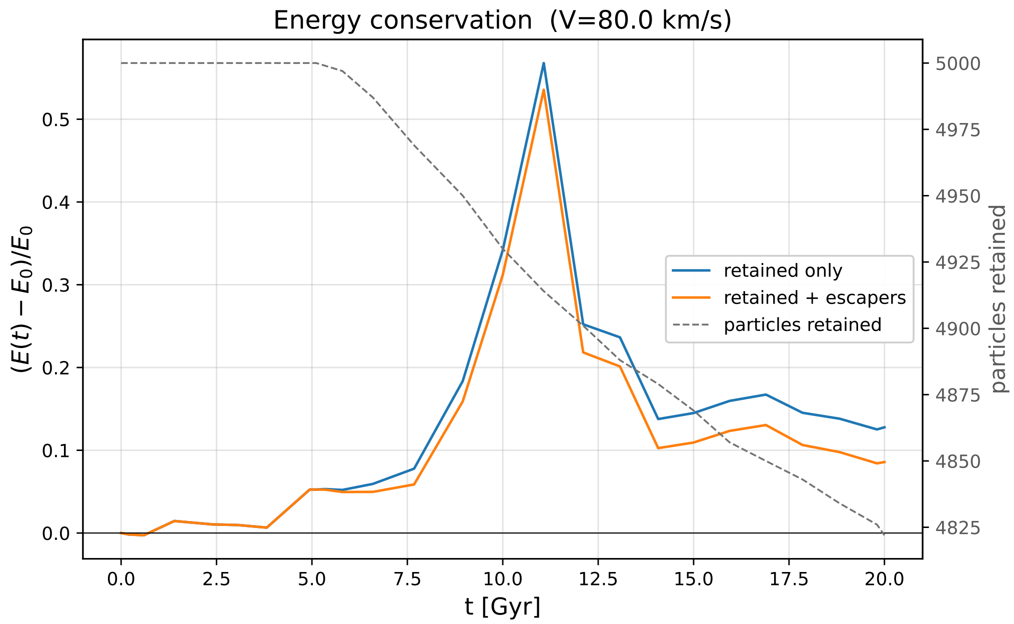

Figure 8: Assigned-tolerance energy and population for $V = 80\,\mathrm{km/s}$, requirement (b).

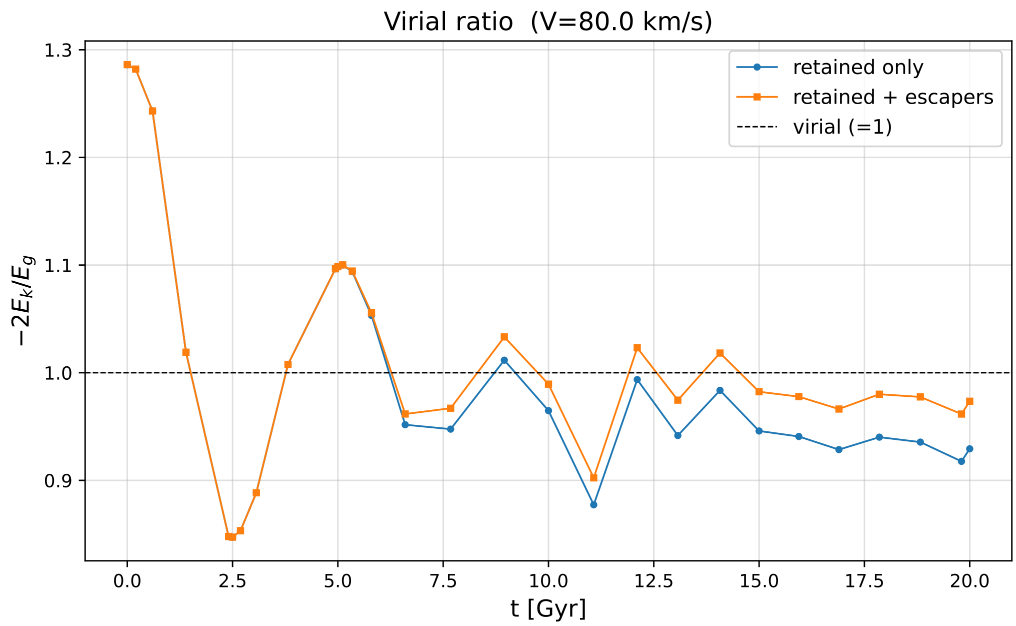

Figure 9: Assigned-tolerance virial ratio for $V = 80\,\mathrm{km/s}$, requirement (c).

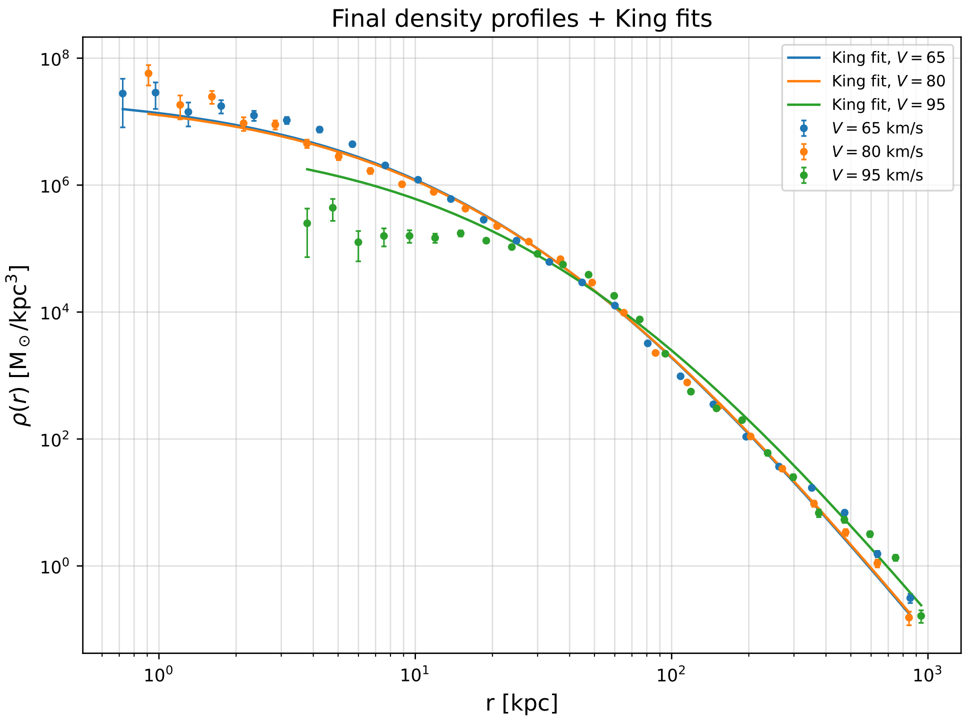

Figure 10: Assigned-tolerance density profiles and King fits, requirement (d).

trend of the converged runs. At 65 km/s, the central density rises from 7.5×10^5 to $2.9 \times 10^7\, M_\odot\, \mathrm{kpc}^{-3}$ because the coarse profile is cuspy. Although the fitted shape changes from $(\alpha, \beta) = (4.831, 0.986)$ to $(0.700, 7.389)$, the constrained product only changes from 4.77 to 5.17. The same integration error removes 245 particles from the bound, contracting 65 km/s system instead of 20 in the converged run.

References

[1] J. Barnes and P. Hut, “A hierarchical $O(N \log N)$ force-calculation algorithm,” *Nature* **324**, 446 to 449 (1986).